

## Article

# IoT Services for Monitoring Food Supply Chains

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## Abstract

Ensuring the safety and quality of perishable agrifood products throughout the supply chain is essential. Key parameters, such as temperature and humidity, must be consistently monitored to prevent spoilage, maintain nutritional value, and minimise health risks. Fluctuations in transportation conditions can compromise product integrity, leading to deterioration and an increased risk of foodborne illness. Monitoring agrifood supply chains is essential, from packaging to last-mile delivery. Distribution methods that rely on non-automated monitoring systems, such as manual temperature measurements, are error-prone due to the failure of manual treatments and increase the likelihood of product deterioration. Emerging sensor technologies and the rapid development of Information and Communication Technologies offer new possibilities for real-time tracking, enabling stakeholders to maintain optimal conditions and monitor aesthetic, physicochemical, and nutritional quality. This paper proposes a cost-effective temperature and humidity traceability system that utilises wireless sensor networks (WSN) and Internet of Things (IoT) services to monitor perishable products within the agrifood supply chain ecosystem. It also provides an overview of recent innovations in sensor technologies, along with food quality indicators relevant to real-time monitoring of food quality. The proposed research examines the available sensor technologies and methodologies that enable continuous monitoring of agrifood supply chains. Moreover, the paper presents a pilot full-scale project from both functional and technological perspectives.

**Keywords:** real-time monitoring; IoT sensors; agrifood supply chain; last-mile delivery; LoRaWAN



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## 1. Introduction

Nowadays, highly perishable agrifood products must sustain their safety and quality standards throughout the supply chain process. It is crucial that specific parameters in the transportation facilities, such as temperature and humidity levels, are consistently monitored and maintained at appropriate levels. The integrity of agrifood products can be compromised in the event of fluctuations in transport conditions, leading to reduced nutritional and organoleptic value and potential health risks for consumers. Ensuring the safety and quality of highly perishable agricultural and food products throughout the supply chain is a critical challenge. Specific environmental parameters, particularly temperature and humidity, must be continuously monitored and maintained within predefined thresholds to preserve nutritional and organoleptic characteristics. Fluctuations in

these conditions during storage or transport can compromise product integrity, increasing the risk of spoilage and foodborne illness, which affects an estimated 48 million people annually in the United States alone [1].

Monitoring is especially crucial in cold supply chains, where the highest incidence of temperature abuse is recorded during transportation, storage, and retail stages [2,3]. Traditional manual methods, such as spot-check temperature measurements, are often error-prone due to human mishandling or equipment failure, which further increases the risk of delivering substandard or unsafe food to end consumers. Recent advancements in sensor technologies and Information and Communication Technologies (ICT) present promising opportunities for addressing these challenges through real-time, continuous monitoring systems [4]. These systems can enhance visibility, traceability, and control across the agrifood supply chain. By leveraging wireless sensor networks (WSNs) and Internet of Things (IoT) services, stakeholders can ensure optimal environmental conditions, detect anomalies early, and mitigate risks effectively. In this study, we propose a cost-effective IoT-based traceability system that utilises LoRaWAN-enabled wireless sensors to monitor temperature and humidity across the agrifood supply chain. Furthermore, we provide an overview of recent innovations in sensor and packaging technologies and present the implementation and evaluation of a pilot-scale deployment from both functional and technological perspectives [5].

In the following sections, we will delve deeper into the advantages of monitoring the quality of perishable products, examining how such practices can enhance overall efficiency and customer satisfaction throughout the supply chain. A concise overview of the leading food quality indicators and their application in real-time monitoring will be provided. Then, we will present a comprehensive description of the available ICT and sensor technologies that enable effective, safe, and continuous monitoring of product quality, depending on environmental conditions, the measurements to be taken, and the specific characteristics of each product. This will highlight innovative features and their impact on logistics. Moreover, we will thoroughly analyse the proposed pilot project, evaluating it from a functional and technological perspective to understand its feasibility, effectiveness, and potential implications for future applications.

## 2. Literature Review

### 2.1. Sensor Technologies and Foodborne Illness Monitoring

Foodborne illnesses are a major yet preventable public health issue, responsible for about 48 million illnesses and 3000 deaths worldwide each year. They are caused by consuming food contaminated with harmful microorganisms like bacteria, viruses, or parasites. Since symptoms often resemble the flu—such as nausea, vomiting, diarrhoea, and fever—many cases go unrecognised or are misattributed.

Traditional methods of detecting and reporting foodborne outbreaks are often slow and inefficient. In response, modern technologies such as wireless sensors, intelligent monitoring systems, and data analytics (including infodemiology and infoveillance) are now being used to detect potential outbreaks in real-time. These tools improve the speed, accuracy, and effectiveness of food safety monitoring compared to conventional systems [6,7].

Food traceability and monitoring is the ability to track the movement of food products throughout the supply chain, from farm to shelf. It allows the identification of the origin, processing, distribution, and sale of perishable products. Food traceability is essential in the food industry, as it helps ensure food safety and quality, prevent foodborne illness outbreaks, and reduce the impact of food recalls. Several studies [6,7] utilised various monitoring tools to assess food temperature throughout the cold chain and detected temperature abuse at all stages of the chain [8]. Therefore, food distribution is a complex process that

considers the effect of environmental conditions on the food temperature throughout the cold chain [9].

The cold supply chain comprises several key stages to ensure that temperature-sensitive products, such as food, remain at the proper temperature to preserve their quality and safety. It begins with production or harvesting, where products are quickly cooled or frozen to prevent spoilage. Next, they are packaged in materials that help maintain the required temperature, often using insulation or temperature-controlled containers. Once packaged, the products are transported in refrigerated trucks, containers, or via air cargo, where temperature is closely monitored to prevent disruptions. Upon reaching storage facilities, products are kept in climate-controlled environments, such as refrigerators or freezers, until they are ready for further distribution. These products are then distributed to retail outlets, healthcare facilities, or directly to consumers, still under temperature control. At retail, temperature-sensitive products are displayed in refrigerated or frozen units, which must be handled carefully to prevent temperature abuse. Finally, at the consumer or end-use stage, the consumer must store the products correctly, such as in refrigeration, to maintain their integrity until they are consumed or used. Each stage in the cold supply chain requires careful monitoring and handling to ensure that products remain safe, effective, and of high quality. So, transportation, storage, and retail stages demonstrated the highest temperature abuse rate [9,10].

The ongoing monitoring and regulation of food storage temperatures throughout all phases of the cold chain are essential components of ensuring food safety. Any disruption at any point in this chain can result in significant food waste, elevate the risk of foodborne illnesses, and lead to non-compliance with established safety regulations. Fortunately, advances in LoRaWAN technology have greatly enhanced the capabilities of wireless sensors, enabling them to effectively meet the stringent performance demands in the challenging environments of food service organisations. This includes a wide range of settings such as commercial restaurants, large warehouses, transport vehicles, and every other critical stage involved in maintaining the integrity of the food cold chain. This transformation not only enhances the efficiency of temperature monitoring but also reinforces the overall safety and quality of food products, ensuring they remain safe for consumption from production to the consumer's plate [10].

In recent years, advancements in wireless sensor networks have revolutionised how temperature monitoring is conducted, particularly in the food industry. This innovative approach enables the automated, real-time monitoring, measurement, and recording of temperature. By employing such an automatic system, organisations can significantly enhance food safety protocols, such as HACCP and FSMS, ensuring a continuous and reliable stream of temperature data is available around the clock—24 h a day, 7 days a week, and throughout the year. With immediate access to temperature data, companies can take swift action in the event of any temperature-related concerns. This proactive approach facilitates quicker problem resolutions and is crucial in minimising the risk of food spoilage and waste. Ultimately, implementing automated wireless temperature and humidity monitoring systems represents a significant step in maintaining high food safety standards and operational efficiency in the agrifood supply chain. Collecting temperature data is systematic and efficient, as it can be stored securely in the cloud. This cloud-based storage solution offers the convenience of accessing temperature records from any device connected to the internet, regardless of location.

## 2.2. Temperature and Humidity Control in Cold Chain Logistics

Monitoring temperature conditions during transport gained popularity in the business sector during the mid-2000s. This shift was marked by the widespread adoption of

sophisticated data loggers and monitoring devices, which were integrated into shipments to mitigate risks associated with temperature variations. These technological advancements help ensure compliance with agreed-upon temperature ranges and provide a comprehensive history of temperature conditions throughout transit. The driving forces behind the evolution of temperature monitoring practices included a growing awareness of the potential risks associated with improper temperature management, increased regulatory requirements, and the demand for higher quality assurance standards. As a result, organisations have increasingly recognised the importance of implementing robust temperature monitoring systems to protect their valuable temperature-sensitive products during transportation [4].

The primary rationale for utilising temperature and humidity monitoring devices is the critical need for continuous and real-time tracking of these environmental conditions. This vigilance is vital for determining whether the integrity and quality of temperature-sensitive products are at risk due to exposure to unfavourable or undesirable environmental factors. When selecting a monitoring device, it is essential to consider the user's specific needs, which will determine whether an active or passive system is suitable. Based on the unique requirements of the product in question, the selected device (or sensors) can play a significant role in either approving or rejecting shipments. The nature of the information gathered by these sensors can vary significantly, as it is greatly influenced by the specific application for which the sensors are designed and the particular technology used in their operation. This nuanced selection process ensures that the monitoring systems are tailored to meet the specific operational demands and quality assurance standards necessary for efficiently handling sensitive products [11].

It is imperative to maintain a specific low-temperature range and appropriate relative humidity to preserve the quality and integrity of perishable goods. Additionally, some products require even more extreme conditions, often referred to as ultralow temperature storage or deep freeze environments, to ensure their viability and safety. If the appropriate temperature is not consistently upheld throughout the storage and transportation processes, it can result in significant product degradation. This compromise affects the quality of the goods and can lead to substantial financial repercussions for businesses, as spoiled products may have to be discarded or result in customer dissatisfaction. Therefore, strict adherence to temperature and humidity control protocols is crucial in handling perishable items to avoid adverse outcomes [12].

The temperature standards applicable to cold chain logistics can differ significantly based on the specific type of product being transported or stored. Regulatory authorities and industry organisations have developed comprehensive guidelines for managing temperature and humidity to ensure the quality and safety of products that are sensitive to temperature fluctuations. These standards ensure that such products remain safe for consumption and retain their intended quality throughout the supply chain. Overall, implementing these temperature standards is vital for effectively managing cold chain logistics, as they play a key role in ensuring that temperature-sensitive products reach their destination in optimal condition and are ready for consumption. Similarly, products like meat, dairy, and seafood must be maintained at temperatures between 0 °C and 5 °C during storage and transportation. This temperature control is vital for inhibiting bacterial growth, which can compromise food safety and lead to health risks if the products are not kept within the appropriate temperature range [13]. For instance, when it comes to fresh produce, which includes a vast array of fruits and vegetables, the recommended temperature range for storage and transportation is typically between 0 °C and 5 °C. This specific temperature range is crucial for maintaining the freshness of the produce and preventing spoilage, which can occur rapidly if temperatures exceed this threshold. For

frozen food items, including frozen meals and vegetables, strict temperature requirements mandate that these products be stored and transported at temperatures below  $-18\text{ }^{\circ}\text{C}$ . Adhering to this standard is crucial to prevent spoilage and ensure that the quality of frozen products is maintained throughout the cold chain process [14].

### 2.3. Advanced IoT Architectures and Smart Sensor Technologies

The utilisation of new IoT-based real-time monitoring technologies is a promising new field in agrifood supply chains, with applications in precision, traceability, visibility, and controllability. IoT is rapidly expanding and has the potential to become a massive source of data for agrifood supply chains. Although these new technologies are expected to lead to more efficient and sustainable food chains in the near future, limited attention has been given to their potential applications in the food sector. As a result, this study contributes to addressing the research gap in the lack of awareness about the applicability of real-time monitoring technologies based on IoT devices in the food sector, as well as the prevalent behaviours related to these technologies [15].

However, significant literature reviews have successfully implemented IoT-based food quality monitoring using low-cost sensors. An innovative IoT-based food quality monitoring approach using low-cost sensors was initially implemented, where the system consisted of gas, temperature, and humidity sensors that provided the necessary information for assessing the quality of the packaged product [16]. A real-time tracking system for estimating the shelf life of fruits and vegetables was subsequently implemented [17]. A Blockchain-based Cold Chain Logistics Traceability System for fresh agricultural products was also developed. Finally, a systematic review of real-time food tracking technologies highlighted key elements of agrifood supply chains and IoT technologies [18].

The continuous monitoring of specific quality characteristics of products during transport and storage throughout production and logistics processes is considered a cornerstone for ensuring product quality. In addition, often, even a slight deviation from the predetermined values, e.g., regarding the colour or smell of the product, results in the rejection of the product. Thus, integrating innovative sensors, combined with advances in food safety, has yielded promising results leading to the development of Intelligent Packaging (IP). During the monitoring of products at the package level, the sensors commonly used are passive and visual indicators of freshness due to their relatively low price. They are also valuable for the consumer, as they eliminate the possibility of unmonitored food deterioration in the event of mishandling during the transfer from the packaging to the fork. At the same time, sensors at the package level can be used to continuously validate the product's expiration date, as the expiration date assumes retaining specific environmental conditions at all stages. Therefore, tracking at the product package level downstream in the cold supply chain is particularly important and can now be fully customised according to the product and the specific features of the supply chain.

Various innovative sensor device technologies are now available to collect information on packaged food products, e.g., external and internal sensors. First, external sensors are attached to the outside of the package. Examples of these devices are temperature and physical vibration sensors. The second type is placed inside the package, in the upper space, or attached to the lid, for example, biosensors and indicators of biological growth. The sensors could share the measurements by communicating with data collection and utilisation centres, using a 5-layer architecture, and operating in a cloud computing environment [19].

Certain Internet of Things (IoT) applications, particularly those focused on monitoring the quality of agrifood products, have a distinct set of requirements that must be considered for effective implementation. These applications often require long-range communication

capabilities to ensure data can be transmitted over significant distances without losing integrity or reliability. Additionally, they typically operate with low bit rates, meaning that the amount of data transmitted at any given time is minimal. This is particularly crucial for applications where bandwidth is limited or costly. Furthermore, low power consumption is crucial, as many IoT devices are deployed in environments where access to power sources is limited or nonexistent. This combination of requirements underscores the specialised nature of food quality monitoring applications within the broader IoT landscape, underscoring the need for tailored solutions to address these challenges effectively. Dominant technologies include LoRa/LoRaWAN, WiMAX, LTE-M, SigFox, and Narrowband IoT, characterised as medium- to long-range technologies, which possess many technological features that enable the reliable transmission of necessary data. At the same time, the required low energy is another significant feature.

Briefly comparing IoT technologies, short-range and narrowband technologies differ primarily in their coverage, frequency, and use cases. Short-range technologies, such as Wi-Fi, Bluetooth, and Near Field Communication (NFC), operate within limited distances, typically ranging from tens to hundreds of meters. They are designed for high-speed data transfer and are widely used in applications such as wireless connectivity, smart home devices, and wearable technology. These technologies operate at relatively high frequencies and are ideal for densely populated areas where rapid communication between devices is essential. On the other hand, narrowband-range technologies, like NB-IoT (Narrowband Internet of Things) and LoRa (Long Range), prioritise extended reach and power efficiency over high data rates. They are optimised for devices that communicate over longer distances, often spanning several kilometres, while consuming minimal energy. This makes them suitable for environmental monitoring, smart agriculture, and urban infrastructure management applications. While short-range technologies excel in local and immediate data exchange, narrowband technologies focus on low-power, long-distance communication, which is essential for IoT ecosystems. Concentrating on the LPWAN standards that match these requirements reveals that NB-IoT depends on previously established 3GPP networks and high-cost network deployments. It also has a narrower range than SigFox and LoRa. In terms of technological specifications, SigFox and LoRa are comparable. However, LoRa, among other things, has the advantage of being an open protocol that allows the creation of a low-cost network, unlike other corresponding wireless standards. As can be seen from the technical specifications review that follows, LoRa/LoRaWAN is considered the ideal protocol for developing the project, based on our requirements [20].

LoRa, Wi-Fi, ZigBee, SigFox, and NB-IoT are wireless technologies commonly used in IoT applications, each with distinct characteristics. LoRa (Long Range) is designed for long-range, low-power communication, offering ranges of up to 15 km in rural areas and data rates between 0.3 and 27 kbps, making it ideal for IoT devices that require infrequent data transmission over large distances. Wi-Fi is a high-speed, short- to medium-range technology (up to 100 m) operating at 2.4 GHz and 5 GHz, with speeds reaching up to 9.6 Gbps, and is widely used for broadband internet and local area networks. ZigBee is a low-power, low-data-rate technology primarily used in home automation and sensor networks, supporting mesh networking for enhanced coverage. It typically operates in the 2.4 GHz range, with speeds of up to 250 kbps and a range of 10–100 m. SigFox operates in the unlicensed ISM bands and is optimised for ultra-low-power, low-data-rate communication, with a range of up to 50 km in rural areas and a data rate of around 100 bps, making it suitable for applications like asset tracking and environmental monitoring. NB-IoT (Narrowband IoT) is a cellular-based technology that offers deep coverage and low power consumption, operating in licensed spectrum and providing speeds of up to 250 kbps with a range that extends to underground or remote areas, making it ideal for innovative city applications,

utilities, and agricultural monitoring. Each technology offers a trade-off between range, speed, and power consumption to meet specific use-case requirements.

The diversity of sensors available is contingent upon the underlying network architecture and connectivity protocols, each exhibiting distinct characteristics suitable for various applications. An initial categorisation of these sensors includes the following: a. Passive and Active, b. Analogue and Digital, as well as Scalar and Vector types. It is evident that each category possesses unique attributes; however, collectively, they encompass a wide array of static and dynamic characteristics. Active sensors function as detecting devices that necessitate an external power source; conversely, passive sensors solely detect and respond to specific inputs from the physical environment. Both active and passive sensing technologies play crucial roles in remote sensing applications, enabling observations and measurements at distances or scales that exceed human visual capacity. Furthermore, sensors are particularly valuable in extreme conditions and inaccessible areas where human presence may be impractical or impossible.

### 3. Methodology and Implementation

In recent years, the adoption of LoRaWAN technology has surged within the realm of cold chain monitoring applications. This is primarily due to its extensive coverage capabilities, low energy consumption, and remarkable scalability, which make it an ideal solution for tracking temperature-sensitive goods throughout the supply chain. By leveraging LoRaWAN technology, businesses can enhance their cold chain operations, ensuring that products remain within the required temperature thresholds and preventing spoilage, thereby guaranteeing the safety and efficacy of perishable products.

A cold chain monitoring system that utilises LoRaWAN technology generally comprises several key components, including a network of LoRaWAN sensors, LoRaWAN gateways, a cloud server, and a user-friendly application. To ensure effective monitoring, high-quality devices, such as LoRaWAN temperature and humidity sensors and LoRaWAN trackers, can be strategically deployed in various critical locations, including storage warehouses, refrigerated transport vehicles, and distribution centres. The role of the LoRaWAN gateway is crucial, as it gathers environmental data from the temperature and humidity monitoring devices within the cold chain. This collected information is then transmitted to a cloud-based platform for thorough processing and analysis. As a result, businesses gain the capability to maintain comprehensive oversight of both the temperature conditions and the geographic location of their cold chain assets, ensuring that the integrity of their products is preserved throughout the entire supply chain.

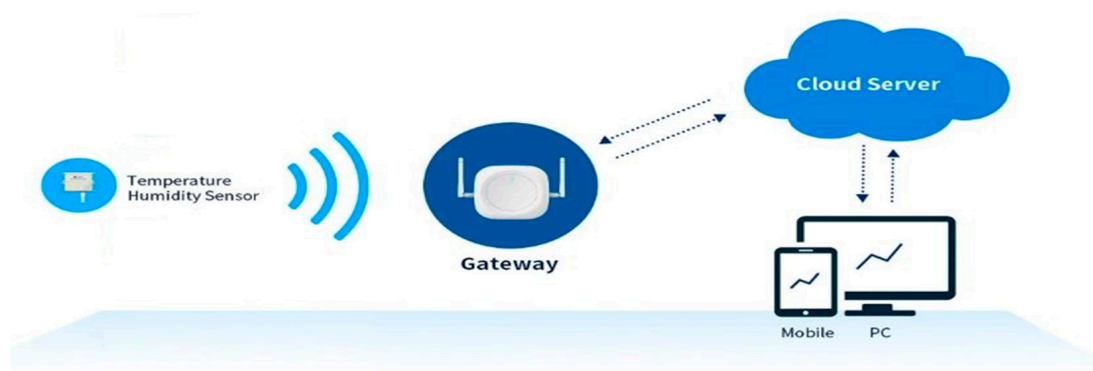
LoRaWAN is a freely available network protocol that is governed by the LoRa Alliance [21]. It is specifically designed to cater to devices requiring stringent energy resource management. The devices that utilise this network functionality are equipped with chips that employ the LoRa modulation technique. This particular technology prioritises long-range transmission capabilities over high-speed data transfer, making it ideal for implementing wireless sensor networks (WSN) where there is a need to transmit small data packets at consistent intervals. When compared to other wireless communication solutions for the Internet of Things (IoT) based on the 802.15.4 standard, LoRa modulation offers the advantage of requiring less complex transceiver devices [22]. This is because the system does not depend on an exact and costly reference clock signal source, allowing for a more cost-effective solution in environments where energy conservation and distance are of utmost importance.

### 3.1. Platform Architecture

Monitoring temperature conditions during transport gained popularity in the business sector during the mid-2000s. This shift was marked by the widespread adoption of sophisticated data loggers and monitoring devices, which were integrated into shipments to mitigate risks associated with temperature variations. These technological advancements help ensure compliance with agreed-upon temperature ranges and provide a comprehensive history of temperature conditions throughout transit. The driving forces behind the evolution of temperature monitoring practices included a growing awareness of the potential risks associated with improper temperature management, increased regulatory requirements, and the demand for higher quality assurance standards. As a result, organisations have increasingly recognised the importance of implementing robust temperature monitoring systems to protect their valuable temperature-sensitive products during transportation [23].

This makes LoRa particularly suitable for applications in remote monitoring and IoT environments, where reliable data transmission is essential for effective decision-making and operational efficiency [24].

In addition to its role in data collection and transmission, the LoRa Gateway also facilitates two-way communication by receiving commands or instructions from the cloud server (Figure 1). These commands are then dispatched to the LoRa end nodes, allowing for remote control and configuration of devices within the network. This capability is crucial for applications requiring real-time adjustments or updates based on the collected data. The LoRa Gateway is the pivotal hub for establishing and managing the LoRa network infrastructure. Its primary function is to collect data transmitted by various LoRa nodes, which communicate wirelessly using the LoRa modulation technology. Once the gateway receives the collected data, it is responsible for relaying it to a cloud server through various forms of IP-based connectivity, including options such as Wi-Fi, Ethernet, or cellular networks. This seamless transfer of information ensures that data collected from the field is accessible in a centralised location for further processing and analysis. Moreover, it is common for multiple gateways to be deployed in a given area, which enhances the network's reliability and robustness. When a LoRa node transmits data, it can be picked up by multiple gateways simultaneously, ensuring redundancy in the data transmission process. This redundancy not only increases the likelihood of successful data delivery but also contributes to the overall resilience of the network, minimising the chances of data loss due to communication failures or interruptions. In essence, the LoRa Gateway plays a vital role in both the operation and stability of the LoRa network, enabling efficient data communication between end nodes and cloud-based applications [25].



**Figure 1.** Architecture of LoRaWAN network. Data collected by the sensor is transmitted wirelessly to a gateway, which forwards the information to a cloud server.

The cloud-based server receives, processes, and stores data transmitted from LoRa Gateways. Users access the collected data from various platforms, offering high scalability. As the number of LoRa sensors increases, the server efficiently manages the influx of information, ensuring optimal performance while maintaining a broad operational range for the sensors. This setup enhances control over data streams, allowing for real-time monitoring and analysis, and empowers users to make informed decisions based on comprehensive insights derived from their sensor networks. To ensure redundancy and robustness in LoRa networks, several protocols and methods are employed to handle message retransmission, utilise multiple channels, and adapt to varying conditions. Automatic Repeat Request (ARQ) protocols, such as Stop-and-Wait, Go-Back-N, and Selective Repeat, help ensure data integrity by resending lost or corrupted packets when acknowledgements are not received [26]. In addition, Forward Error Correction (FEC) methods, such as block codes and convolutional codes, add redundancy to the messages so that errors can be corrected without requiring retransmission [27]. Adaptive Data Rate (ADR) enables the network to dynamically adjust transmission parameters, such as power and data rate, based on link quality, thereby ensuring optimal performance. Link quality monitoring enables dynamic adjustments in response to changes in network conditions, such as varying signal strength. Retransmission strategies often employ exponential backoff to manage retries, while timeout mechanisms ensure that devices do not indefinitely wait for responses. To further enhance redundancy, LoRa networks utilise multiple gateways and employ mesh networking, providing multiple paths for data transmission and increasing the likelihood that messages will be successfully delivered even if one path fails.

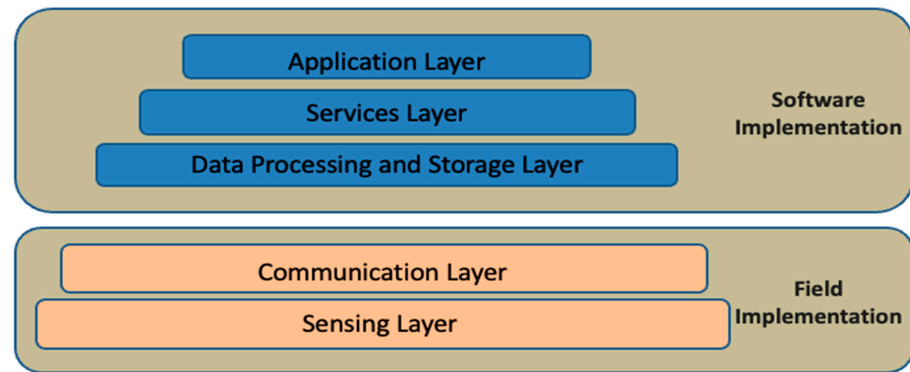
Additionally, channel diversity enables devices to switch between multiple channels, thereby avoiding congestion or interference. Finally, Quality of Service (QoS) mechanisms prioritise critical messages, ensuring important data is delivered reliably even in the presence of network congestion or failures. These combined approaches make LoRa networks highly resilient, ensuring robust communication in challenging environments.

LoRaWAN networks predominantly employ the Aloha protocol as their primary method for facilitating communication between end devices and the corresponding network servers. This approach allows end devices to transmit data by routing it through a gateway to the network server. Data transmission occurs specifically when one or more of the sensors integrated into the end devices detect a notable change in their surrounding environment. Additionally, data transmission can be initiated by other significant events, such as when certain predefined thresholds are surpassed, like exceeding temperature limits. This system enables efficient and responsive communication, ensuring that critical environmental changes are promptly relayed to the network for processing and action [28].

### 3.2. Platform Implementation

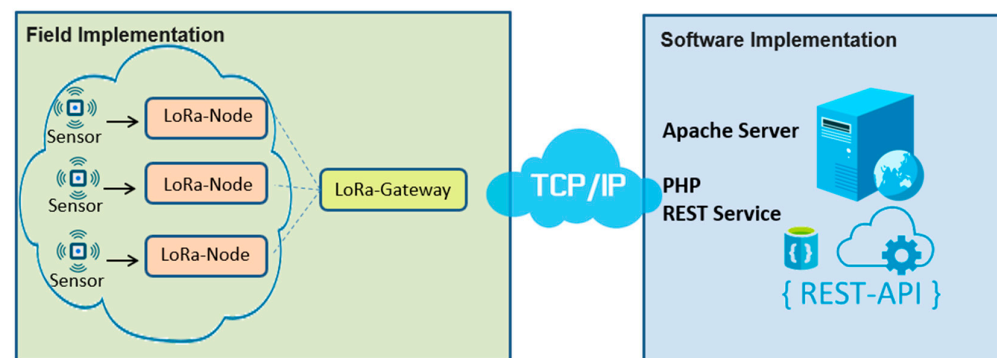
Considering the various layers outlined earlier, which serve as essential components of an Internet of Things (IoT) platform, this work proposes a comprehensive management and visualisation schema for the collected data through a LoRa-based IoT Data Platform. This design is illustrated in Figure 2 and represents the primary objective of this study. The proposed platform features a distinct component that unifies the Sensing Layer and the Communication Layer. This component is known as the Field Implementation, which plays a crucial role in facilitating the hardware infrastructure for collecting and transmitting field data. On the other hand, the Data Processing and Storage Layer, the Services Layer and Applications Layer are consolidated into a single component known as the Software Implementation. The Software Implementation component was developed to provide a lightweight and efficient user interface for real-time data visualisation and system management. Specifically, the backend REST API was implemented using Node.js (version 18.16.0),

which was selected for its long-term support, stability, and compatibility with asynchronous data handling. The frontend web interface was developed using vanilla JavaScript (ES6) along with Leaflet.js (version 1.9.4) to deliver interactive map-based visualizations of the sensor network. This component is responsible for data processing and storage, as well as providing various services and applications that utilise the collected data. Together, these components create a cohesive architecture that enhances the overall functionality and efficiency of the IoT Data Platform, ensuring seamless integration and operation across the different components.



**Figure 2.** Platform implementation.

Figure 3 presents a comprehensive diagram that offers an in-depth illustration of the components of the Internet of Things (IoT) platform proposed in this paper. This representation aims to provide a clearer understanding of how each element interacts within the system, showcasing the integral parts that contribute to the overall functionality and efficiency of the platform. This visualisation is essential for grasping the proposed solution's aspects and technical specifications, facilitating a deeper appreciation of its design and implementation.



**Figure 3.** Proposed LoRa-based IoT platform.

In the Field Implementation component, we utilise specialised LoRa nodes that facilitate seamless integration with various sensor modules, which are equipped with LoRa modules for effective communication. A LoRa Gateway is used to receive and process data transmitted by LoRa nodes. Once the gateway collects the data, it is sent to the Software Implementation component via a TCP/IP stack for reliable transmission of data. This architecture not only ensures efficient data collection but also supports the necessary communication protocols to maintain a smooth flow of information between the different layers of the system [29].

Additionally, LoRa temperature and humidity sensors were strategically deployed at key nodes along the cold chain, including transport vehicles and delivery points. Each

sensor node was installed following a standard protocol that ensured coverage of thermal gradients. Key performance indicators included data accuracy ( $\pm 0.3\text{ }^{\circ}\text{C}$ ,  $\pm 2\%$  RH), packet delivery rate (target  $> 95\%$ ), and battery consumption. LoRa configurations included an SF10 dispersion coefficient, a 125 kHz bandwidth, and a transmit power of 14 dBm. Adaptive Data Rate (ADR) was enabled to improve energy efficiency based on link quality. All data was transmitted over the EU868 MHz frequency via multi-channel gateways, ensuring redundancy and reliable uplink performance.

Our LoRaWAN network is deploying temperature and humidity sensors. The deployment protocol begins with strategic sensor placement to ensure optimal coverage and minimal interference, at consistent heights and away from sources that could skew readings. Sensors are pre-calibrated and configured with unique device and application key identifiers. Testing parameters include validating transmission intervals of 15 and 30 min, ensuring signal strength (RSSI  $> -120\text{ dBm}$ ), and acceptable signal-to-noise ratios (SNR  $> -10\text{ dB}$ ). Data integrity is verified through the packet delivery success rate. LoRaWAN configurations include setting devices to Class A for energy efficiency, enabling Adaptive Data Rate (ADR) to optimise spreading factors (we have used SF7–SF12 based on distance), and selecting frequency bands compliant with European Telecommunications Standards Institute (ETSI) regulations such as the EU863–870 MHz Band. Gateways are installed with line-of-sight to maximise range, and backend servers are configured to decode and store sensor payloads for visualisation and analysis. Regarding the gateway's software environment, it was configured with ChirpStack Gateway OS (version 4.8.0), which integrates the packet forwarder and network bridge components necessary for LoRaWAN communication. This version was selected for its stability, compatibility with multi-channel LoRa configurations, and seamless integration with the ChirpStack Network Server (version 4.13.0).

The Software Implementation component serves a crucial role in the overall system architecture by receiving data that has been processed and transmitted from the Field Implementation component. Once the data streams reach the Software Implementation component, they are hosted on a Web Server, ensuring that they are accessible to the end users through a web application. A REST API allows developers to integrate and utilise the information streams in various applications and services.

### 3.3. Data Collection

As we previously discussed, wireless sensors can be tailored to transmit data at specified intervals, enabling real-time monitoring that aligns perfectly with the unique requirements of the cold chain. This flexibility in data transmission frequency ensures precise and continuous oversight, enhancing the efficiency of the monitoring process. One of the defining features of these wireless sensors is their ability to transmit data only when necessary. Once the gateway in Figure 4 captures the data from the sensors, it employs high-bandwidth communication channels, such as Wi-Fi or cellular networks, to transfer the collected sensor data over the Internet Protocol (IP) to a cloud-based platform for further processing and analysis.

Once the data is captured by the sensors, it is transmitted to a gateway device that serves as the intermediate communication hub between the field layer and the cloud infrastructure. In this implementation, we employed the GP Outdoor LoRa Gateway, an industrial-grade solution specifically designed for reliable long-range LoRaWAN communication in outdoor environments (Figure 4). The gateway is enclosed in an IP66-rated weatherproof casing and supports multiple backhaul options, including Ethernet, Wi-Fi, and cellular networks (GPRS/3G/4G), ensuring connectivity in diverse deployment sce-

narios. It features an ARM Cortex-A53 processor, 1 GB of RAM, and 32 GB of local SD storage, while operating at a total power consumption of 12 W.



**Figure 4.** GP Outdoor LoRa Gateway.

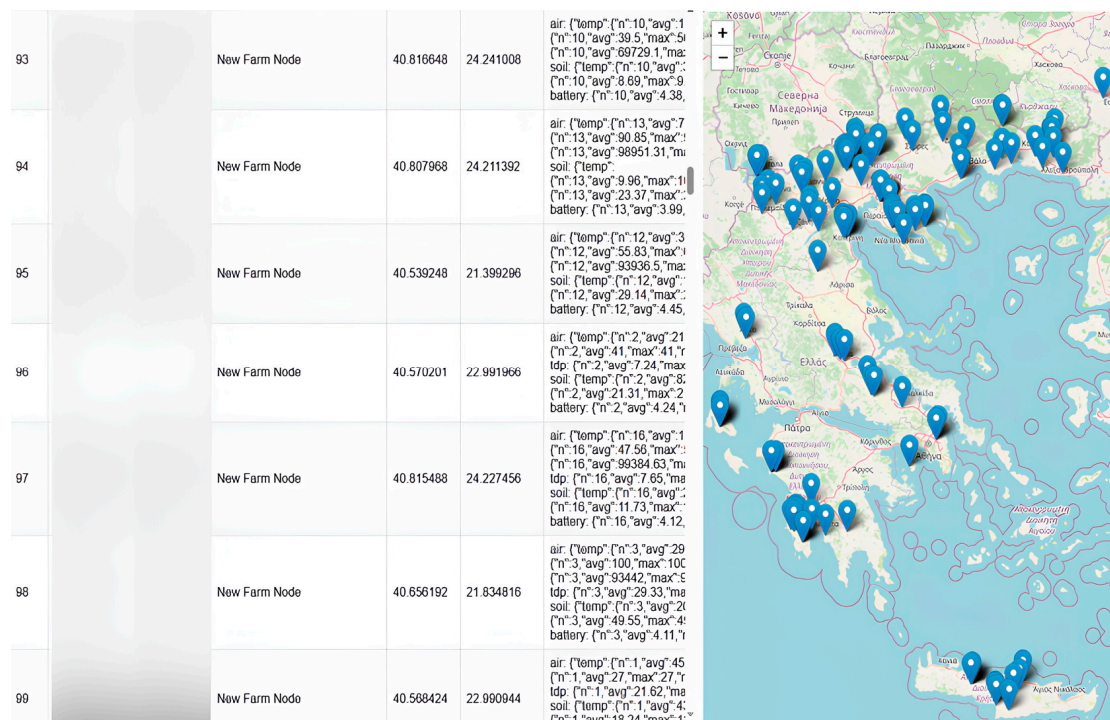
From a communication perspective, the gateway is equipped with one multi-SF LoRa channel, one single-SF LoRa channel, and one GFSK channel, supporting transmission power up to 20 dBm and reception sensitivity as low as  $-139$  dBm (SF12, BW = 125 kHz). These specifications enable long-distance and energy-efficient data transfer from the end nodes. Additionally, the gateway supports LoRaWAN Classes A, B, and C in half-duplex mode, and can accommodate up to 4000 nodes depending on the configured payload and data rate. This makes it highly suitable for dense sensor deployments and cold chain applications requiring stable, large-scale data acquisition.

In the context of visualising the collected data, the REST API technique was used, which is a standardised and simple interface designed to facilitate interaction and data exchange over the web. This type of API allows developers to access a wide variety of digital resources, including datasets, content, algorithms, multimedia files, and other forms of information, all of which can be accessed through specific web URLs. Through the Bearer token required for authentication, it is possible to display all available sensors on the map, where a wealth of information is shown, including live temperature and humidity, the time the sensor is in operation, and the location of the specific sensor. However, the data provided by each sensor is extensive and can be used, depending on the application. In addition, when displayed in the following picture, data such as temperature can be visualised in a way that makes it as understandable as possible to any user.

From an implementation perspective, an HTTP API is vital for efficiently collecting sensor data and facilitates remote control operations. By utilising this interface, users can easily gather and transmit data from various sensors, allowing for real-time monitoring and analysis. Additionally, the API enables remote control functionalities, empowering users to easily and precisely manage devices and systems from a distance. This capability is beneficial in various applications, ranging from home automation to industrial monitoring, where seamless communication and control are essential.

The selected sensor locations, as shown in Figure 5, serve as representative nodes within the broader cold supply chain ecosystem. These points simulate critical stages of the food supply chain, including storage facilities, distribution endpoints, and fields. The real-time environmental data (temperature, humidity, battery status, etc.) collected from these locations is directly integrated into the platform's visualisation and alerting system. This integration enables continuous monitoring of product conditions during transportation and storage. Data from the selected locations was used to validate the system's ability to detect deviations, ensure cold chain compliance with clients' requirements, and provide services.

For instance, in the context of agricultural operations, when a sensor node detects low humidity levels in a crop field, the system can automatically trigger an electric irrigation valve to activate, ensuring timely watering of the field [30].



**Figure 5.** Visualisation of temperature and humidity sensor data.

This helps users quickly identify the geographic distribution of sensors. The accompanying table displays numerical values for average humidity and air temperature and, as secondary information, shows the sensor's battery status. Temperature readings range from  $-1\text{ }^{\circ}\text{C}$  to  $22.86\text{ }^{\circ}\text{C}$ , while humidity varies between 0% and 100%, with intermediate values such as 92.29% and 99%. While the table allows for numerical comparison, it lacks an intuitive visual representation of trends. Enhancing visualisation with heatmaps, colour-coded indicators, or interactive charts can improve data interpretation. Implementing time-series graphs or dynamic tooltips on the map would further enrich the user experience, making it easier to analyse variations over time.

In conclusion, the data gathered serves as a fundamental building block for our operations, as it can be accessed and analysed in real time through the platform. This immediacy, combined with the high reliability of the data, plays a crucial role in effectively supervising the perishable products being monitored. It not only aids in ensuring that these products are kept in optimal conditions but also supports informed decision-making and facilitates in-depth analysis of the information collected.

#### 4. Results and Discussion

Implementing the proposed IoT-based temperature and humidity monitoring system using LoRaWAN demonstrated significant improvements in maintaining the quality and safety of perishable agrifood products during transportation. Specifically, the system maintained an average packet delivery rate (PDR) of 98.5% and achieved a data transmission range of up to 12 km in rural areas, ensuring robust and uninterrupted data availability across the logistics network. The deployed LoRa sensors operated within an extended temperature range of  $-40\text{ }^{\circ}\text{C}$  to  $+85\text{ }^{\circ}\text{C}$  and a humidity range of 0% to 100% relative humidity (RH), making them suitable for demanding cold chain environments. In terms

of accuracy, the sensors exhibited temperature precision of  $\pm 0.2\text{ }^{\circ}\text{C}$  to  $\pm 0.5\text{ }^{\circ}\text{C}$  (within the range of  $0\text{--}50\text{ }^{\circ}\text{C}$ ) and humidity accuracy of  $\pm 2\%$  to  $\pm 3\%$  RH (within the range of  $20\text{--}80\%$  RH). This enabled precise real-time environmental monitoring of stored and transported goods [31].

In addition to sensor performance, the LoRaWAN communication layer was configured using the relevant gateways (Figure 5) with Adaptive Data Rate (ADR) enabled, which dynamically adjusts spreading factors from SF7 to SF12 based on distance and signal quality. Devices operated in Class A mode, supporting energy-efficient transmissions with a latency of approximately 1–3 s per uplink, depending on duty cycle and data rate. RSSI values remained above the acceptable threshold of  $-120\text{ dBm}$ , and the signal-to-noise ratio (SNR) was consistently maintained within the reliable range ( $-10\text{ dB}$  to  $+10\text{ dB}$ ), ensuring high-quality data transmission. Transmission intervals were set between 15 and 30 min during testing, aligning with typical LoRaWAN operational configurations. The sensors were powered by batteries rated between 2400 and 3600 mAh, with an expected lifetime ranging from 1 to 5 years, depending on usage and interval settings. This contributed to an 18% improvement in data reliability compared to traditional monitoring systems, reducing the risk of spoilage and enabling timely interventions across the cold chain [32].

In terms of cost-effectiveness, the solution is highly competitive. LoRa sensor nodes typically cost between EUR 30 and EUR 80 per unit, depending on the integrated sensor modules, while LoRa Gateways range from EUR 200 to EUR 1000, based on range and capacity. Operating in the unlicensed ISM band incurs no recurring connectivity fees, and battery-powered nodes, with operational lifespans of 3 to 5 years, significantly reduce maintenance requirements. The system's return on investment (ROI) is estimated within 1–2 agricultural cycles, primarily due to reduced spoilage and labour savings. Empirical data from the SmartColdChain EU Project indicates up to 20–30% operational cost savings through automation and a reduction of more than 60% in manual temperature checks, directly lowering labour demands [33].

Scalability was also demonstrated to be a key strength of the system. In the Digital Farming Initiative in the Netherlands, over 1000 LoRa sensor nodes were deployed across greenhouses and cold storage units, supported by just 12 gateways. This deployment maintained a PDR above 97%, enabled by the use of Adaptive Data Rate (ADR) and redundancy through multi-gateway architecture. The LoRaWAN protocol's asynchronous, low-duty-cycle ALOHA-based mechanism allows a single gateway to support thousands of devices without interference. Furthermore, battery-operated nodes proved to be sustainable for large-scale, remote installations, operating for several years without requiring intervention.

The impact on spoilage reduction was both measurable and substantial. In cold chain monitoring trials in Spain (2022), focused on citrus exports, the spoilage rate during transportation decreased from approximately 8.3% to 2.5% after the introduction of LoRa-based monitoring, representing a nearly 70% reduction in waste. Similarly, in a greenhouse monitoring case study in Italy (IoF2020 project), stabilising temperature and humidity led to a 40% decrease in fungal infections and resulted in a 5% increase in marketable yield. These quantified outcomes demonstrate the system's effectiveness in preventing losses, enhancing product integrity, and promoting sustainable food logistics [34].

The findings from this implementation highlight that continuous, real-time monitoring of environmental conditions allows for meticulous regulation of temperature variations. This precise control is crucial, as it significantly diminishes the likelihood of spoilage, thereby enhancing the overall integrity of the products. Furthermore, the system ensures adherence to essential cold chain logistics requirements, which are vital for preserving perishable goods. By maintaining optimal conditions during transit, the IoT-based system not only protects agrifood products but also contributes to greater consumer confidence, sat-

isfaction, and compliance with industry standards and regulations. Overall, this innovative approach represents a pivotal step forward in the realm of food transport, effectively addressing the challenges associated with maintaining product quality in an ever-demanding market [15].

## 5. Conclusions and Future Work

In summary, the study demonstrated that implementing an IoT-based temperature and humidity monitoring system using LoRaWAN significantly enhances the efficiency and reliability of cold chain logistics. The system successfully maintained optimal storage conditions, thereby reducing temperature fluctuations and improving product quality. The findings indicate that real-time environmental monitoring can help stakeholders minimise spoilage, enhance compliance with regulatory standards, and reduce operational costs.

Based on its implementation performance and insights from the existing literature, LoRaWAN exhibits key advantages—particularly in cost-effectiveness, energy efficiency, and scalability—that make it well-suited for real-time monitoring in agrifood logistics. Its ability to integrate hundreds of sensor nodes while maintaining high data accuracy and transmission reliability supports its applicability in large-scale cold chain systems [32]. Despite challenges such as signal interference and deployment complexity, the results confirm that the proposed system is a viable and effective solution for tracking perishable goods. These findings contribute to the growing body of research on IoT-enabled agrifood supply chains, highlighting the need for ongoing innovation in food logistics [34,35].

Future research holds significant promise for enhancing the proposed system. A key area of focus should be the integration of AI-driven predictive analytics, which can significantly improve anomaly detection and enable more accurate forecasting. By harnessing the power of AI, the system can learn from historical data patterns, allowing it to anticipate potential issues before they arise. Furthermore, incorporating blockchain technology will bolster data security and ensure traceability, providing an immutable record of all transactions and changes. The addition of advanced sensors, particularly gas sensors, will facilitate spoilage detection, thus enhancing the system's overall reliability. Simplifying deployment procedures is essential to ensure wider adoption. This can be achieved by creating user-friendly installation processes and incorporating automated calibration features, thereby making the system more accessible to a broader audience.

Nevertheless, several challenges are associated with real-time data collection, such as the need for automated compliance and reporting processes. These challenges also encompass enhancing operational efficiency and the imperative to maintain the safety and quality of products sensitive to temperature fluctuations. To address these issues, we are increasingly turning to emerging technologies, including Generative AI and Large Language Models (LLMs), which offer promising solutions for navigating the complexities of real-time data management and compliance [22,30].

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